Fixed Effects - 3 categories

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In this analysis, we will model relationships for the type of Spanish produced v/s the social and/or demographic variables and sentence type. Sentence type variable, due to the large number of categories and limited data due to 6 speakers, was very sparse and it was difficult to fit any type of model reliably with the current structure. So, I reduced the number of sentence types to sentence groups/ categories based on how similar some statements were, this grouping may not defensible from a linguistic stand point.

The new grouping for sentence type is: 1. Sentence Type 1-6 : Statements 2. Sentence Type 7-11: yes/ no questions 3. Sentence Type 12-14: wh- questions 4. Sentence type >= 15: commands or others

The response has 3 categories: - 1: Spanish - 2: Ecuador - 3: Different

The goal of the analysis is to answer the following questions: 1. Did Ecuadorian speakers learn to produce Spanish patterns?

2. Did Ecuadorian speakers still produce/use Spanish patters in Ecuador?
3. In Ecuador, which patterns (Spain or Ecuador) are more produced/used? and determine which factors (age, level of education, gender, time of stay in Spain or return to Ecuador, AND sentence type) promote the use/production of either or?
4. What are the different patterns? intermediate forms between Spain and Ecuador patterns? patterns from other dialects of Spanish? errors?

Since we want to see whether Spanish patterns are learned in the first 2 questions, we will treat Ecuador (2) as our baseline level and then model the relationship to make interpretations easier. So model coefficients can be used to say things like how likely Spanish or "Different Patterns" are over Ecuador patterns.

In order to answer the first 2 questions we will use predicted probabilities and construct confidence intervals around them using bootstrapping. We will construct 95% confidence intervals.

Answering the 3rd question is not easy since every variable has a different scale, so we can't just compare coefficients, so we will look at the predicted probabilities for different values of the variable under study while averaging over the other predictors. For e.g. if we wanna see how important age is we look at predicted probability of age = 29 v/s probability of age = 44.

```r
library(readxl)
df <- read_excel("data_returnees_3categories.xlsx")
head(df)
```

```
## # A tibble: 6 x 8
##   Speaker   age gender stay_spain return_ecuador edu_level sentence_type    dv
##   <chr>   <dbl>  <dbl>      <dbl>          <dbl>     <dbl>         <dbl> <dbl>
## 1 S01        29      1         16          0.417         3             1     3
## 2 S01        29      1         16          0.417         3             1     3
## 3 S01        29      1         16          0.417         3             1     3
## 4 S01        29      1         16          0.417         3             1     3
## 5 S01        29      1         16          0.417         3             1     3
## 6 S01        29      1         16          0.417         3             1     3
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
df <- df[complete.cases(df), ]
df_model <- df %>%
  mutate(
    dv = factor(dv,
                levels = c(1, 2, 3),
                labels = c("Ecuador", "Spain", "Different")),
    dv = relevel(dv, ref = "Ecuador"),
    Speaker = factor(Speaker),
    gender = factor(gender),
    edu_level = factor(edu_level),
    sentence_type = as.numeric(as.character(sentence_type)),
    sent_group = case_when(
      sentence_type %in% 1:6 ~ "statement",
      sentence_type %in% 7:11 ~ "yn_question",
      sentence_type %in% 12:14 ~ "wh_question",
      sentence_type %in% 15:17 ~ "command_other",
      TRUE ~ NA_character_
    ),
    sent_group = factor(sent_group,
                        levels = c("statement", "yn_question", "wh_question", "command_other"))
  ) %>%
  filter(!is.na(sent_group))
```

Assuming independence within speaker observations for now (probably a wrong assumption):

```r
library(nnet)

fit_fixed <- multinom(
  dv ~ sent_group + age + gender + stay_spain + return_ecuador + edu_level,
  data = df_model,
  Hess = TRUE,
  trace = FALSE
)

summary(fit_fixed)
```

```
## Call:
## multinom(formula = dv ~ sent_group + age + gender + stay_spain +
##     return_ecuador + edu_level, data = df_model, Hess = TRUE,
##     trace = FALSE)
##
## Coefficients:
##             (Intercept) sent_groupyn_question sent_groupwh_question
```

```
## Spain      -0.6175499             0.8708167             0.4044543
## Different  -0.1929455            -0.2184265            -1.5069224
##           sent_groupcommand_other         age   gender1 stay_spain
## Spain                -0.005939069 -0.00293931 0.7273033 0.05924039
## Different            -2.464988219  0.01770379 0.4557797 0.08603007
##           return_ecuador edu_level3 edu_level4
## Spain        -1.66519094 -0.7387629 -0.7603455
## Different     0.05490415 -0.7981191 -0.5220470
##
## Std. Errors:
##           (Intercept) sent_groupyn_question sent_groupwh_question
## Spain       0.2497483             0.4623796             0.4765683
## Different   0.2317555             0.3623924             0.4150293
##           sent_groupcommand_other        age   gender1 stay_spain
## Spain                   0.4373369 0.01781216 0.4011965 0.06051102
## Different               0.4145593 0.01637584 0.3514352 0.05553398
##           return_ecuador edu_level3 edu_level4
## Spain          0.8871479  0.6402577  0.4318467
## Different      0.8534478  0.6153217  0.4049998
##
## Residual Deviance: 653.6394
## AIC: 689.6394
```

Model metrics:

```r
pred_class <- predict(fit_fixed, type = "class")
pred_prob  <- predict(fit_fixed, type = "probs")

table(observed = df_model$dv, predicted = pred_class)
```

```
##            predicted
## observed    Ecuador Spain Different
##   Ecuador        57    10        43
##   Spain          26    21        41
##   Different      17    11       118
```

```r
summary(pred_prob)
```

```
##     Ecuador             Spain             Different
##  Min.   :0.1306   Min.   :0.04996   Min.   :0.1066
##  1st Qu.:0.1780   1st Qu.:0.12620   1st Qu.:0.2126
##  Median :0.2917   Median :0.26673   Median :0.4428
##  Mean   :0.3198   Mean   :0.25581   Mean   :0.4244
##  3rd Qu.:0.4244   3rd Qu.:0.38364   3rd Qu.:0.5834
##  Max.   :0.7557   Max.   :0.44604   Max.   :0.7408
```

```r
apply(pred_prob, 2, range)
```

```
##        Ecuador      Spain Different
## [1,] 0.1306321 0.04995906 0.1065530
## [2,] 0.7557369 0.44603900 0.7407781
```

For confidence intervals:

```r
coefs <- summary(fit_fixed)$coefficients
ses   <- summary(fit_fixed)$standard.errors
```

```
lower <- coefs - 1.96 * ses
upper <- coefs + 1.96 * ses
```

Probability of Spanish Patterns overall, based on the model:

```
pred_prob <- as.data.frame(predict(fit_fixed, newdata = df_model, type = "probs"))

pred_df <- cbind(df_model, pred_prob)

mean_spain <- mean(pred_df$Spain)

mean_spain
```

```
## [1] 0.2558143
```

Do they produce Spanish patterns or learn to produce Spanish patterns overall: (Confidence interval computed using bootstrapped samples.)

```
library(boot)
```

```
## Warning: package 'boot' was built under R version 4.5.3
```

```
boot_fun <- function(data, indices) {
  d <- data[indices, ]

  fit <- multinom(
    dv ~ sent_group + age + gender + stay_spain + return_ecuador + edu_level,
    data = d,
    trace = FALSE
  )

  pred <- as.data.frame(predict(fit, newdata = d, type = "probs"))
  mean(pred$Spain)
}

set.seed(123)

boot_out <- boot(
  data = df_model,
  statistic = boot_fun,
  R = 1000
)

boot_out$t0
```

```
## [1] 0.2558143
```

```
quantile(boot_out$t, c(0.025, 0.975), na.rm = TRUE)
```

```
##      2.5%     97.5%
## 0.2093056 0.3023256
```

Based on Stay in Spain and Return to Ecuador, we will compute similar confidence intervals for predicted probability

```
library(ggeffects)
```

```
## Warning: package 'ggeffects' was built under R version 4.5.3
```

```r
pred_stay <- ggpredict(
  fit_fixed,
  terms = "stay_spain [all]"
)

pred_return <- ggpredict(
  fit_fixed,
  terms = "return_ecuador [all]"
)

pred_stay
```

```
## # Predicted probabilities of dv
##
## dv: Ecuador
##
## stay_spain | Predicted
## ----------------------
##          5 |      0.26
##         11 |      0.18
##         12 |      0.17
##         13 |      0.16
##         16 |      0.13
##
## dv: Spain
##
## stay_spain | Predicted
## ----------------------
##          5 |      0.12
##         11 |      0.11
##         12 |      0.11
##         13 |      0.11
##         16 |      0.11
##
## dv: Different
##
## stay_spain | Predicted
## ----------------------
##          5 |      0.62
##         11 |      0.71
##         12 |      0.72
##         13 |      0.73
##         16 |      0.77
##
## Adjusted for:
## *     sent_group = statement
## *            age =     34.75
## *         gender =         0
## * return_ecuador =      0.23
## *      edu_level =         2
```

```r
pred_return
```

```
## # Predicted probabilities of dv
```

```
##
## dv: Ecuador
##
## return_ecuador | Predicted
## --------------------------
##           0.03 |      0.17
##           0.05 |      0.17
##           0.06 |      0.17
##           0.17 |      0.17
##           0.42 |      0.18
##           0.67 |      0.18
##
## dv: Spain
##
## return_ecuador | Predicted
## --------------------------
##           0.03 |      0.15
##           0.05 |      0.15
##           0.06 |      0.15
##           0.17 |      0.12
##           0.42 |      0.08
##           0.67 |      0.06
##
## dv: Different
##
## return_ecuador | Predicted
## --------------------------
##           0.03 |      0.68
##           0.05 |      0.69
##           0.06 |      0.69
##           0.17 |      0.71
##           0.42 |      0.74
##           0.67 |      0.76
##
## Adjusted for:
## * sent_group = statement
## *        age =     34.75
## *     gender =         0
## * stay_spain =     11.47
## *  edu_level =         2
```

```
prob_spain_at_boot <- function(data, indices, a, b) {
  d <- data[indices, ]

  fit <- multinom(
    dv ~ sent_group + age + gender + stay_spain + return_ecuador + edu_level,
    data = d,
    trace = FALSE
  )

  newdata <- d
  newdata$stay_spain <- a
  newdata$return_ecuador <- b
```

```r
  pred <- as.data.frame(predict(fit, newdata = newdata, type = "probs"))
  mean(pred$Spain)
}
```

```r
get_spain_ci <- function(data, a, b, R = 1000) {
  set.seed(123)

  boot_out <- boot(
    data = data,
    statistic = function(data, indices) prob_spain_at_boot(data, indices, a, b),
    R = R
  )

  c(
    estimate = boot_out$t0,
    lower = quantile(boot_out$t, 0.025, na.rm = TRUE),
    upper = quantile(boot_out$t, 0.975, na.rm = TRUE)
  )
}
get_spain_ci(df_model, a = 12, b = 0.0493, R = 500)
```

```
##    estimate  lower.2.5% upper.97.5%
##   0.3061168   0.2403349   0.3794389
```

```r
grid_df <- df_model %>% distinct(stay_spain, return_ecuador)
out <- t(mapply(
  FUN = function(a, b) {
    get_spain_ci(df_model, a = a, b = b, R = 300)
  },
  a = grid_df$stay_spain,
  b = grid_df$return_ecuador
))

out <- as.data.frame(out)
names(out) <- c("spain_prob", "spain_lower", "spain_upper")

grid_df <- cbind(grid_df, out)
grid_df
```

```
##   stay_spain return_ecuador spain_prob spain_lower spain_upper
## 1         16         0.4167  0.2038479   0.1109140   0.2970336
## 2         12         0.0493  0.3061168   0.2462968   0.3752333
## 3          5         0.1667  0.2455217   0.1365528   0.3978431
## 4         11         0.0274  0.3107624   0.2395666   0.3803371
## 5         12         0.6667  0.1405202   0.0603327   0.2403503
## 6         13         0.0575  0.3058454   0.2434680   0.3810117
```

Similar thing for different sent groups:

```r
prob_spain_group_boot <- function(data, indices, g) {
  d <- data[indices, ]

  fit <- multinom(
    dv ~ sent_group + age + gender + stay_spain + return_ecuador + edu_level,
    data = d,
    trace = FALSE
```

```r
  )

  newdata <- d
  newdata$sent_group <- factor(g, levels = levels(d$sent_group))

  pred <- as.data.frame(predict(fit, newdata = newdata, type = "probs"))
  mean(pred$Spain)
}

get_spain_group_ci <- function(data, g, R = 1000) {
  set.seed(123)

  boot_out <- boot(
    data = data,
    statistic = function(data, indices) {
      prob_spain_group_boot(data, indices, g)
    },
    R = R
  )

  c(
    estimate = boot_out$t0,
    lower = quantile(boot_out$t, 0.025, na.rm = TRUE),
    upper = quantile(boot_out$t, 0.975, na.rm = TRUE)
  )
}

group_df <- data.frame(
  sent_group = levels(df_model$sent_group)
)

out <- t(sapply(group_df$sent_group, function(g) {
  get_spain_group_ci(df_model, g = g, R = 500)
}))

out <- as.data.frame(out)
names(out) <- c("spain_prob", "spain_lower", "spain_upper")

group_df <- cbind(group_df, out)
group_df
```

```
##                  sent_group spain_prob spain_lower spain_upper
## statement         statement  0.1261639  0.07184321   0.1933560
## yn_question     yn_question  0.2836732  0.19395782   0.3778560
## wh_question     wh_question  0.3418862  0.22181745   0.4476017
## command_other command_other  0.3211502  0.22519653   0.4198317
```

Comparing different variables spain probabilities:

```r
library(ggeffects)

ggeffect(fit_fixed, terms = "sent_group")
```

```
## # Predicted probabilities of dv
##
```

```
## dv: Ecuador
##
## sent_group    | Predicted |     95% CI
## --------------------------------------
## statement     |      0.21 | 0.14, 0.30
## yn_question   |      0.20 | 0.13, 0.29
## wh_question   |      0.39 | 0.27, 0.52
## command_other |      0.54 | 0.43, 0.65
##
## dv: Spain
##
## sent_group    | Predicted |     95% CI
## --------------------------------------
## statement     |      0.12 | 0.07, 0.20
## yn_question   |      0.28 | 0.20, 0.38
## wh_question   |      0.33 | 0.22, 0.47
## command_other |      0.31 | 0.22, 0.42
##
## dv: Different
##
## sent_group    | Predicted |     95% CI
## --------------------------------------
## statement     |      0.67 | 0.57, 0.76
## yn_question   |      0.52 | 0.42, 0.62
## wh_question   |      0.28 | 0.18, 0.41
## command_other |      0.15 | 0.09, 0.24
```

```r
pred_sent <- ggeffect(fit_fixed, terms = "sent_group")
pred_sent
```

```
## # Predicted probabilities of dv
##
## dv: Ecuador
##
## sent_group    | Predicted |     95% CI
## --------------------------------------
## statement     |      0.21 | 0.14, 0.30
## yn_question   |      0.20 | 0.13, 0.29
## wh_question   |      0.39 | 0.27, 0.52
## command_other |      0.54 | 0.43, 0.65
##
## dv: Spain
##
## sent_group    | Predicted |     95% CI
## --------------------------------------
## statement     |      0.12 | 0.07, 0.20
## yn_question   |      0.28 | 0.20, 0.38
## wh_question   |      0.33 | 0.22, 0.47
## command_other |      0.31 | 0.22, 0.42
##
## dv: Different
##
## sent_group    | Predicted |     95% CI
## --------------------------------------
## statement     |      0.67 | 0.57, 0.76
```

```
## yn_question    |      0.52 | 0.42, 0.62
## wh_question    |      0.28 | 0.18, 0.41
## command_other |      0.15 | 0.09, 0.24
```

```r
pred_age <- ggeffect(fit_fixed, terms = "age")
pred_age
```

```
## # Predicted probabilities of dv
##
## dv: Ecuador
##
## age | Predicted |     95% CI
## ----------------------------
##  26 |      0.34 | 0.27, 0.42
##  29 |      0.34 | 0.28, 0.40
##  34 |      0.33 | 0.27, 0.39
##  35 |      0.33 | 0.27, 0.39
##  41 |      0.31 | 0.24, 0.39
##  44 |      0.31 | 0.23, 0.40
##
## dv: Spain
##
## age | Predicted |     95% CI
## ----------------------------
##  26 |      0.28 | 0.22, 0.35
##  29 |      0.27 | 0.22, 0.33
##  34 |      0.26 | 0.21, 0.32
##  35 |      0.26 | 0.21, 0.31
##  41 |      0.24 | 0.18, 0.32
##  44 |      0.23 | 0.16, 0.32
##
## dv: Different
##
## age | Predicted |     95% CI
## ----------------------------
##  26 |      0.38 | 0.31, 0.46
##  29 |      0.39 | 0.33, 0.46
##  34 |      0.41 | 0.35, 0.47
##  35 |      0.42 | 0.36, 0.48
##  41 |      0.45 | 0.37, 0.53
##  44 |      0.46 | 0.37, 0.56
```

```r
pred_stay_spain <- ggeffect(fit_fixed, terms = "stay_spain")
pred_stay_spain
```

```
## # Predicted probabilities of dv
##
## dv: Ecuador
##
## stay_spain | Predicted |     95% CI
## -----------------------------------
##          5 |      0.44 | 0.29, 0.61
##         11 |      0.33 | 0.28, 0.39
##         12 |      0.32 | 0.26, 0.38
##         13 |      0.30 | 0.24, 0.37
```

```
##         16 |      0.26 | 0.17, 0.37
##
## dv: Spain
##
## stay_spain | Predicted |     95% CI
## -----------------------------------
##          5 |      0.24 | 0.13, 0.39
##         11 |      0.26 | 0.20, 0.31
##         12 |      0.26 | 0.21, 0.32
##         13 |      0.26 | 0.20, 0.32
##         16 |      0.26 | 0.17, 0.38
##
## dv: Different
##
## stay_spain | Predicted |     95% CI
## -----------------------------------
##          5 |      0.32 | 0.20, 0.48
##         11 |      0.41 | 0.35, 0.47
##         12 |      0.42 | 0.36, 0.49
##         13 |      0.44 | 0.37, 0.51
##         16 |      0.48 | 0.36, 0.60
```

```r
pred_gender <- ggeffect(fit_fixed, terms = "gender")
pred_gender
```

```
## # Predicted probabilities of dv
##
## dv: Ecuador
##
## gender | Predicted |     95% CI
## -------------------------------
## 0      |      0.39 | 0.30, 0.49
## 1      |      0.27 | 0.19, 0.36
##
## dv: Spain
##
## gender | Predicted |     95% CI
## -------------------------------
## 0      |      0.21 | 0.15, 0.30
## 1      |      0.30 | 0.22, 0.40
##
## dv: Different
##
## gender | Predicted |     95% CI
## -------------------------------
## 0      |      0.40 | 0.31, 0.49
## 1      |      0.43 | 0.34, 0.53
```

```r
pred_return_ecuador <- ggeffect(fit_fixed, terms = "return_ecuador")
pred_return_ecuador
```

```
## # Predicted probabilities of dv
##
## dv: Ecuador
##
```

```
## return_ecuador | Predicted |     95% CI
## ---------------------------------------
##           0.03 |      0.30 | 0.22, 0.38
##           0.05 |      0.30 | 0.23, 0.38
##           0.06 |      0.30 | 0.23, 0.38
##           0.17 |      0.32 | 0.26, 0.38
##           0.42 |      0.35 | 0.27, 0.44
##           0.67 |      0.37 | 0.22, 0.55
##
## dv: Spain
##
## return_ecuador | Predicted |     95% CI
## ---------------------------------------
##           0.03 |      0.33 | 0.25, 0.41
##           0.05 |      0.32 | 0.25, 0.40
##           0.06 |      0.32 | 0.25, 0.40
##           0.17 |      0.28 | 0.23, 0.34
##           0.42 |      0.20 | 0.14, 0.28
##           0.67 |      0.14 | 0.07, 0.26
##
## dv: Different
##
## return_ecuador | Predicted |     95% CI
## ---------------------------------------
##           0.03 |      0.38 | 0.29, 0.47
##           0.05 |      0.38 | 0.30, 0.47
##           0.06 |      0.38 | 0.30, 0.47
##           0.17 |      0.40 | 0.34, 0.47
##           0.42 |      0.45 | 0.36, 0.54
##           0.67 |      0.49 | 0.32, 0.66
```

Tried fitting a mixed effects model, but wasn't successful due to lack of unique data, the design matrix is computationally singular, which is a problem and the algorithm doesn't converge. Although, mixed effects should be used, it isn't possible to do so right now since we don't have enough data. A solution could be to use a Bayesian model instead with a weakly informative prior(s), but that requires more rigorous research, so skipping that for now. The most ideal solution would be to talk to more speakers and get a more representative data. Inferences should be made carefully, there is a lot of scope for improvement.